1960 SUPPLEMENT* TO FEDERAL SPECIFICATION

LAMPS; ELECTRIC INCANDESCENT MINIATURE, TUNGSTEN-FILAMENT

This supplement was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies, and shall become effective not later than September 1, 1959. It may be put into effect, however, at any earlier date after promulgation by the Administrator, General Services Administration.

I. TABLES OF INITIAL TOLERANCES

TABLE I.—*Tungsten-filament lamps for flashlight service*

Trade No.	Initial rating tolerance at design volts (amperes variation in percent, plus or minus)	Light center length tolerances	Axial alinement tolerances	Multiplying factor for basic life tolerances [1]
		Inch	*Inch*	
PR6	7	0.010	0.010	2
PR7	7	.010	.010	2
13	7	--------	--------	2
14	7	--------	--------	2
31	7	--------	--------	2
222	7	--------	--------	2
243	7	--------	--------	2
385	7	--------	--------	2

[1] The life tolerance for any type of lamp is determined by multiplying the basic life tolerance for the number tested by the factor given above.

*The supplement to the Federal specification for miniature lamps is amended or issued annually to correspond with the contract period, and is an integral part of the specification.

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TABLE III.—*Miniature tungsten-filament lamps for aviation service*

Trade No.	Initial rating tolerances at design volts			Light center length tolerances	Axial alinement tolerances	Multiplying factor for basic life tolerances
	Variation in candles per watt, plus or minus	Variation in spherical candle-power, plus or minus	Variation in amperes, plus or minus			
	Percent	*Percent*	*Percent*	*Inch*	*Inch*	
301	15	20	10	3/32	3/32	2.0
302	15	20	10	3/32	3/32	2.0
303	15	20	10	3/32	3/32	2.0
305	12	15	8	1/16	1/16	1.5
307	12	15	8	1/16	1/16	1.5
308	12	15	8	1/16	1/16	1.5
309	10	12	8	3/64	3/64	1.2
311	10	12	7	3/64	3/64	1.2
313	15	----------	10	3/32	3/32	2.0
327	----------	50	10	----------	----------	2.0
328	----------	----------	10	----------	----------	----------
330	----------	----------	15	----------	----------	----------

TABLE IV.—*Miniature tungsten-filament lamps for special service*

Trade No.	Initial rating tolerances at design volts			Multiplying factors for basic life tolerances
	Variation in candles per watt, plus or minus	Variation in spherical candle-power, plus or minus	Variation in amperes, plus or minus	
	Percent	*Percent*	*Percent*	
63K	6	8	5	1.5
67K	12	15	8	2.0
81K	7	9	5	1.5

II. SCHEDULES OF COMMERCIAL RATINGS OF MINIATURE TUNGSTEN-FILAMENT LAMPS

SCHEDULE 1.—*Miniature tungsten-filament lamps for flashlight service*

Trade No.	Design volts	Amperes	Base	Bulb	Maximum overall length	Light center length	Rated average laboratory life
					Inches	*Inch*	*Hours*
PR6	2.47	0.30	Min. flange S.C.	B-3½	1¾	* ¼	30
PR7	3.70	.30	do	B-3½	1¾	* ¼	30
13	3.70	.30	Min. screw S.C.	G-3½	1 5/16	2 3/32	15
14	2.47	.30	do	G-3½	1 5/16	2 3/32	15
31	6.15	.30	do	G-4½	1 11/16	2 3/32	15
222	2.25	.25	do	TL-3	1 5/16	----------	5
243	2.33	.27	do	TL-3	1 5/16	----------	10
365	3.69	.50	do	G-3½	1 5/16	2 3/32	15

* Light center length is measured from top of locating bosses.

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SCHEDULE 2.—*Miniature tungsten-filament lamps for automotive service*

LAMPS WITH 2-PIN BAYONET BASES

Trade No.	Circuit volts	Design volts	Design amperes	Rated spherical candle-power	Base	Bulb	Maximum overall length	Light center length	Initial candles (spherical) per watt	Candle-power maintenance at ⅓ life [1]	Rated average laboratory life
							Inches	*Inches*		*Percent of initial*	*Hours*
51	6	7.50	0.22	1	Min S.C.	G-3½	1 5/16	½	0.60		1,000
55	6	7.00	.41	2	do	G-4½	1 1/16	½	.70		500
57	12	14.00	.24	2	do	G-4½	1 1/16	½	.60		500
63	6	7.00	.63	3	Cand. S.C.	G-6	1 7/16	¾	.68	91	1,000
64	6	7.00	.63	3	Cand. D.C.	G-6	1 7/16	¾	.68	91	1,000
67	12	13.50	.59	4	Cand. S.C.	G-6	1 7/16	1 3/16	.50		(2 4)
68	12	13.50	.59	4	Cand. D.C.	G-6	1 7/16	1 3/16	.50		(2 4)
82	6	6.50	1.02	6	do	G-6	1 7/16	¾	.90	75	500
87	6	6.75	1.91	15	Cand. S.C.	S-8	2	1⅛	1.16	85	300
88	6	6.75	1.91	15	Cand. D.C.	S-8	2	1⅛	1.16	85	300
89	12	13.00	.58	6	Cand. S.C.	G-6	1 7/16	¾	.80	82	750
90	12	13.00	.58	6	Cand. D.C.	G-6	1 7/16	¾	.80	82	750
623	24	28	.37	6	do	G-6	1 7/16	¾	.58		1,000
1000	6	6.20	3.87	{32} {32}	Cand. D.C.	RP-11	2¼	1¼	1.33	[3] 94	125
1004	12	12.8	.94	15	do	B-6	1¾	1 1/16	1.25	75	100
1034	12	12.8	1.80	32	D.C. Indexing	S-8	2	1¼	1.39	[3] 80	200
		14.0	.51	4					.56		(4)
1073	12	12.8	1.80	32	Cand. S.C.	S-8	2	1¼	1.39	80	200
1129	6	6.40	2.63	21	Cand. S.C.	S-8	2	1¼	1.25	86	200
1133	6	6.20	3.91	32	do	RP-11	2¼	1¼	1.32	92	200
1141	12	12.80	1.34	21	do	S-8	2	1¼	1.22	80	500
1142	12	12.80	1.34	21	Cand. D.C.	S-8	2	1¼	1.22	80	500
1154	6	6.40	2.63	21	D.C. Indexing	S-8	2	1¼	1.25	[3] 85	200
		7.00	.75	3					.57		1000
1158	6	6.40	2.63	21	Cand. D.C.	S-8	2	1¼	1.25	[3] 85	200
		7.00	.75	3					.57		1000
1203	24	28	.71	21	S.C. bay	S-8	2	1¼	1.06		400
1251	24	28	.23	3	do	G-6	1 7/16	¾	.47		1,000
1683	24	28	1.02	32	do	S-8	2	1¼	1.12		500

[1] These maintenance values apply to base-horizontal burning.
[2] At 14.0 volts.
[3] Figure applies for life test on first filament.
[4] Indefinitely long—over 2,000 hours.

TABLE II.—*Miniature tungsten-filament lamps for automotive service*

LAMPS WITH 2-PIN BAYONET BASES

Trade No.	Initial rating tolerances at design volts			Light center length tolerance	Axial alinement tolerance from planes B and C	Multiplying factor for basic life tolerances
	Variation in candles per watt, plus or minus	Variation in spherical candlepower, plus or minus	Variation in amperes, plus or minus			
	Percent	*Percent*	*Percent*	*Inch*	*Inch*	
51	15	20	10	3/32	3/32	2.0
55	10	12	7	3/32	3/32	1.8
57	15	20	10	3/32	3/32	2.0
63 and 64	6	8	5	3/32	3/32	1.5
67 and 68	12	15	8	3/32	3/32	
82	7	9	5	3/32	3/32	1.5
87 and 88	7	9	5	1/16	1/16	1.0
89 and 90	12	15	8	3/32	3/32	1.5
623	15	20	10	3/32	3/32	2.0
1000 [1]	5	7	4	0.04	[2] 0.03	1.0
1004 [1]	7	9	5	1/16	1/16	1.5
1034: [1]						
Major filament	8	10	5	.04	.04	1.5
Minor filament	10	12	8			
1073 [1]	8	10	5	.04	.04	1.5
1129 [1]	5	7	4	.04	.04	1.0
1133 [1]	5	7	4	.04	.04	1.0
1141 [1]	9	10	6	.04	.04	1.0
1142 [1]	9	10	6	.04	.04	1.0
1154:						
Major filament	6	8	5	.04	.04	1.0
Minor filament	10	12	8			1.5
1158: [1]						
Major filament	6	8	5	.04	.04	1.0
Minor filament	10	12	8			1.5
1203	12	15	8	3/64	3/64	1.5
1251	15	20	10	3/32	3/32	2.0
1683	12	15	8	3/64	3/64	2.0

[1] Angular variation of base pins for lamps. A line through the contact points of the base locking pins shall be at an angle of 0°±3° with reference to planes A and B (par. F-8) and at an angle of 0°±15° with a plane through the centers of the light sources and normal to plane A (except lamp No. 1158 for which this angle is 90°±15°).

[2] Neither light source exclusive of filament loop shall have a greater length, parallel to the base axis, than 0.12 inch. The light sources shall be approximately parallel to each other and their centers shall be 0.14±0.015 inch apart. The center of each light source shall be 0.07±0.03 inch from reference plane C (par. F-8).

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SCHEDULE 3.—*Miniature tungsten-filament lamps for aviation service*

Trade No.	Circuit volts	Design volts	Design amperes	Rated spherical candlepower	Base	Bulb	Maximum overall length	Light center length	Initial candles (spherical) per watt	Rated average laboratory life	Filament
							Inches	*Inches*		*Hours*	
301	28	28.0	0.17	3	Cand. bay, S.C.	G-5	1⅜	1 1/16	0.62	500	C-2F
302	28	28.0	.17	3	Cand. bay, D.C.	G-5	1⅜	1 1/16	.62	500	C-2F
303	28	28.0	.31	6	Cand. bay, S.C.	G-6	1 7/16	¾	.70	500	C-2F
305	28	28.0	.53	15	do	S-8	2	1⅛	1.01	300	C-2V
307	28	28.0	.67	21	do	S-8	2	1⅛	1.12	300	C-2V
308	28	28.0	.67	21	Cand. bay, D.C.	S-8	2	1⅛	1.12	300	C-2V
309	28	28.0	.90	32	Cand. bay, S.C.	S-11	2⅜	1¼	1.27	300	C-2V
311	28	28.0	1.29	50	do	S-11	2⅜	1¼	1.39	300	C-2V
313	28	28.0	.17		Min. bay, S.C.	T-3¼	1 3/16	5/8	.73	500	C-2F
327	28	28.0	.04	0.33	Midget flange, S.C.	T-1¾	5/8	3/8		1,000	C-21 or C-2F
328		6.0	.20		do	T-1¾	5/8	3/8		500	C-6
330	14	14.0	.08		do	T-1¾	5/8	3/8		750	C-2V or C-2F

SCHEDULE 4.—*Miniature tungsten-filament lamps for special service*

Trade No.	Circuit volts	Design volts	Design amperes	Nominal spherical candlepower	Base	Bulb	Maximum overall length	Light center length	Initial candles (spherical) per watt	Candlepower maintenance at 1/5 life	Rated average laboratory life
							Inches	*Inches*		*Percent of initial*	*Hours*
63K	7	7.00	0.63	3	Cand. screw S.C.	G-6	1⅜	1 5/16	0.73	91	1,000
67K	13.50	13.50	.44	4	do	G-6	1⅜	1	.70	83	[1] 1,000
81K	6.5	6.50	1.02	6	do	G-6	1⅜	1 5/16	.99	75	500

[1] At 14.0 volts.

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